Optimal Experimental Designs for Wiener Process Models in Clinical Research Studies

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Abstract

We consider disease progression models formulated as Wiener processes and develop new approaches for optimizing design of clinical studies that employ such models. We focus on a model that jointly characterizes the mean drift, within-process variance, and between-subject (population) variance. Covariate effects on these parameters are modeled through a generalized regression framework, yielding a flexible class of parametric models. Within this framework, various optimal design problems can be formulated and solved using established theoretical tools. The proposed methodology is illustrated through optimal design problems motivated by randomized controlled trials for disease-modifying therapies.

Keywords: Clinical trials, disease progression, Wiener process.

1 Introduction

There has been a considerable research on degradation processes in various scientific fields [1–3]. Degradation models have been widely studied in reliability engineering where the primary motivation is to assess the lifetime of highly reliable products that do not fail frequently under standard conditions [4]. An important area of engineering applications is accelerated degradation testing, where higher than normal stress levels (e.g., temperature, voltage) are applied to experimental units (e.g., long-life products) to estimate various parameters characterizing the reliability of the products [5–7].

While in biomedical research the use of degradation models is less common, some relevant methodologies and applications can be found in [8–11]. Applying reliability engineering's stochastic frameworks for degradation models in clinical research studies poses several challenges. The human biology is very complex and an individual disease progression can be influenced by various factors, such as genetics, demographic characteristics, environmental factors, among others. Clinical studies typically have limited duration and relatively short follow-up periods that may not fully capture long-term degradation patterns. Also, for many diseases there is a lack of well-established biomarkers that could enable accurate prediction of individual disease trajectory accounting for relevant covariates and uncertainties.

For many progressive diseases, there is a strong need to develop disease-modifying treatments that may delay, slow down, or halt disease progression. Optimal designs of clinical trials for generating robust evidence that experimental treatments alter the course of a disease compared to standard-of-care treatments warrant investigation.

This paper focuses on disease progression models based on a Wiener process (also known as Brownian motion process with drift) and suggests methods for optimizing design of clinical trials of disease-modifying therapies. Section 2 formulates the model for disease progression in clinical study settings. Section 3 derives the elemental Fisher information matrix for model parameters (including the case when these parameters may depend on covariates) and formulates an optimal design problem for a clinical trial with Wiener process-based outcomes. Section 4 considers some special cases and examples showcasing an application of the proposed methodology. Section 5 concludes with a discussion and outlines future work.

2 Wiener's model for disease progression and data acquisition model

2.1 Disease progression model

Let the disease progression be described by a random process $Y(t)$ with a drift parameter μ. In the simplest case, $Y(t)$ can be approximated with a solution of the linear stochastic differential equation

$$\begin{aligned} &dY(t) = \mu dt + \sigma_W dW(t), \\ &Y(t_0) = y_0, \quad W(t_0) = 0, \quad \text{and} \quad t_0 \le t \le t^*, \end{aligned} \tag{1}$$

where $W(t)$ is a standard Wiener process ($W(0) = 0$, $W(t+u) - W(t) \sim \mathcal{N}(0, u)$ for $u > 0$), $[t_0, t^*]$ is the interval over which the assumed working model is reasonable, y_0 is an initial (i.e., at moment t_0) level of degradation, and $\sigma_W > 0$ is the scale (volatility) parameter [8, 12, 13]. A solution to (1) is

$$Y(t) = y_0 + \mu(t - t_0) + \sigma_W W(t - t_0). \tag{2}$$

$Y(t)$ is normally distributed with mean $y_0 + \mu(t - t_0)$ and variance $\sigma_W^2(t - t_0)$. The statistical characterization of the process (2) at any $t > t_0$ is entirely determined by y_0 and t_0. This implies that for parameter estimation and future state prediction, there is no requirement to identify the exact moment of the disease onset, as the current baseline value y_0 effectively captures all necessary information from the past to build valid estimators for the drift μ and volatility σ_W.

In real-life settings, (2) can be viewed as an asymptotic version of the finite difference equation

$$Y(t_j + \Delta) - Y(t_j) = \mu\Delta + \sigma_W W(\Delta), \quad t_j = t_0 + j\Delta,\ j = 0, 1, \ldots, n, \text{ and } n\Delta = t^*,$$

with either $\Delta \to 0$ for a fixed $t^* > 0$ or $t^* \to \infty$ for a fixed $\Delta > 0$.

In many applications, (2) is observed until the first occurrence of a value of $Y(t)$ equal to or greater than a given threshold $\gamma > 0$, in which case the observations are stopped, and the respective time T is recorded as "failure time." If $Y(t = T) = \gamma$, then T is random and has the inverse Gaussian distribution [2, 14] with probability density function

$$f(t) = \frac{\gamma - y_0}{\sqrt{2\pi\sigma_W^2(t - t_0)^3}} \exp\left\{-\frac{[(\gamma - y_0) - \mu(t - t_0)]^2}{2\sigma_W^2(t - t_0)}\right\},\ t > t_0,$$

which has mean $\mathrm{E}[T] = t_0 + \frac{\gamma - y_0}{\mu}$ and variance $\mathrm{Var}[T] = \frac{(\gamma - y_0)\sigma_W^2}{\mu^3}$. Also, $\mathrm{E}[T^{-1}] = \frac{\mu}{\gamma - y_0} + \frac{\sigma_W^2}{(\gamma - y_0)^2}$ and $\mathrm{Var}[T^{-1}] = \frac{\mu\sigma_W^2}{(\gamma - y_0)^3} + \frac{2\sigma_W^4}{(\gamma - y_0)^4}$.

In clinical research, this stochastic framework enables the prediction of the survival function $S(t) = \Pr(T > t | \gamma, \mu, \sigma_W^2)$ while the majority of study participants still remain below the failure threshold γ. By estimating the mean drift (μ) and volatility (σ_W^2) early in the trial—whether through a simple plug-in method or a using a mixture of inverse Gaussian distributions—researchers can forecast long-term outcomes without requiring direct observation of failure times (T). This approach provides significant ethical and operational value, as it allows for the assessment of treatment efficacy and safety without necessitating the prolonged exposure of patients to potentially ineffective or unsafe therapies.

The process (2) typically describes within-subject variability. Its applicability can be extended by applying some popular transformations like $G(t) = e^{Y(t)} > 0$ or $G(t) = L + \frac{U - L}{1 + e^{-Y(t)}}$, where L and U are some constants, $L < U$, such that $L \leq G(t) \leq U$.

The drift μ may depend on some covariates $\mathbf{x}$. In the presence of substantial between-subject variability, one may introduce the random μ, assuming, for instance, that it is normally distributed. Similar considerations can be applied for the parameter σ_W. Further, instead of a linear equation (1), stochastic differential equations of higher order may be considered. Finally, in some applications, the health condition of a patient is described with several potentially correlated biomarkers, in which case one may consider multivariate extensions of (2).

2.2 Statistical modeling in clinical study settings

In clinical research studies, an individual participant (study patient) is monitored over some pre-determined period $[t_0, t^*]$ during which observations on $Y(t)$ are acquired at time points $0 \leq t_0 < t_1 < \cdots < t_n = t^*$. Typically the initial moments t_0 are different across study patients and are defined by the enrollment process.

Two types of individual disease progression trajectories should be distinguished. The first type corresponds to the case when a patient's trajectory was observed as $Y(t)$ for $t \in [t_0, t^*]$, and all these observations were below the failure threshold γ. The second type corresponds to the case when the patient's trajectory was observed at time points $\{t_0, t_1, \ldots, t_l\}$ for some $l < n$, and all these observations were below γ. Then the trajectory $Y(t)$ crossed the threshold $(Y(t) \geq \gamma)$ at some random time point T, where $t_l < T \leq t^*$. The patient's outcome was recorded as "failure" at $t = T$, and no further data for the patient was collected.

In this paper, for simplicity, we assume that the value of γ is sufficiently large such that $\Pr\left\{\max_{t\in[t_0,t^*]} Y(t) < \gamma\right\}$ is close to 1, and the data collection time points are fixed (non-stochastic). Therefore, we focus on the trajectories of the first type.

For the ith study patient $(i = 1, \ldots, N)$, n_i observations on the disease progression $Y_i(t)$ are collected at time points t_{ij}, where $j = 0, 1, \ldots, n_i$. The ith patient's baseline is denoted as $y_{i0} = Y_i(t_{i0})$, and it is assumed to be known. Subsequent observations on the same patient will be denoted by $Y_{ij} = Y_i(t_{ij})$. In heterogeneous populations, the drift may be specific for each patient, in which case it will be denoted by μ_i. Therefore, for the ith patient, the "measurement" model can be written as

$$Y_{ij} = Y_{i(j-1)} + \mu_i(t_{ij} - t_{i(j-1)}) + \sigma_W W_i(t_{ij} - t_{i(j-1)}), \tag{3}$$

where μ_i is assumed to be normally distributed with $\mathrm{E}[\mu_i] = \mu$ and $\mathrm{Var}[\mu_i] = \sigma_\mu^2$, and the parameters μ and σ_W correspond to the disease progression model (2).

In general, the sequences $\{t_{ij}\}_{j=0}^{n_i}$ are patient-specific. This may occur by design in controlled experiments (e.g., to gain more information given the administrative, medical, and cost constraints) and/or at random (e.g., in observational studies or in clinical trials where some measurements from a patient may be missing). In the former case, the "unblinding" problem may potentially arise. In the case of missing data, some statistical issues due to "missingness not at random" (MNAR) may arise.

Let $\mathbf{V}_i^\top = \{Y_{ij} - Y_{i(j-1)}\}_{j=1}^{n_i}$, $\boldsymbol{\tau}_i^\top = \{\tau_{ij}\}_{j=1}^{n_i} = \{t_{ij} - t_{i(j-1)}\}_{j=1}^{n_i}$, and $\mathbf{W}_i^\top = \{W_i(\tau_{ij})\}_{j=1}^{n_i}$. Then (3) can be written in the matrix form as

$$\mathbf{V}_i = \mu_i \boldsymbol{\tau}_i + \sigma_W \mathbf{W}_i, \tag{4}$$

and

$$\mathrm{E}[\mathbf{V}_i] = \mu\boldsymbol{\tau}_i = \boldsymbol{\mu}_i \quad \text{and} \quad \boldsymbol{\Sigma}_i = \mathrm{Var}[\mathbf{V}_i] = \sigma_\mu^2 \boldsymbol{\tau}_i\boldsymbol{\tau}_i^\top + \sigma_W^2 \mathbf{S}_i, \tag{5}$$

where $\mathbf{S}_i = \mathrm{diag}\{\tau_{ij}\}_{j=1}^{n_i}$ is an $n_i \times n_i$ diagonal matrix. Throughout the paper, the vector $\boldsymbol{\tau}_i$ will be referred to as "data acquisition protocol" (DAP).

3 Setting the study design problem

3.1 Elemental Fisher information matrix

In optimal design theory, the information gained from a single patient ("information unit") is a crucial component. In this section, we will use the results from the theory of multivariate normal observations [15, 16] to derive the Fisher Information Matrix (FIM) for the considered models. For notation simplicity, we suppress the patient index i when no ambiguity arises.

Let $\boldsymbol{\zeta}$ ($\dim \boldsymbol{\zeta} = m'$) be the vector of parameters to be estimated. Let $\boldsymbol{\mu}(\boldsymbol{\zeta})$ and $\boldsymbol{\Sigma}(\boldsymbol{\zeta})$ denote the mean and the covariance matrix, respectively. Then the (α, β) element of the FIM for $\boldsymbol{\zeta}$ is given by

$$\mathscr{M}_{\alpha\beta}(\boldsymbol{\zeta}) = \frac{\partial \boldsymbol{\mu}^\top(\boldsymbol{\zeta})}{\partial \zeta_\alpha} \boldsymbol{\Sigma}^{-1}(\boldsymbol{\zeta}) \frac{\partial \boldsymbol{\mu}(\boldsymbol{\zeta})}{\partial \zeta_\beta} + \frac{1}{2}\,\mathrm{tr}\left[\boldsymbol{\Sigma}^{-1}(\boldsymbol{\zeta})\frac{\partial \boldsymbol{\Sigma}(\boldsymbol{\zeta})}{\partial \zeta_\alpha}\boldsymbol{\Sigma}^{-1}(\boldsymbol{\zeta})\frac{\partial \boldsymbol{\Sigma}(\boldsymbol{\zeta})}{\partial \zeta_\beta}\right]; \tag{6}$$

see, for instance, Theorem 15.6 in [17]. Following [15], the matrix $\mathscr{M} = \{\mathscr{M}_{\alpha\beta}(\boldsymbol{\zeta})\}$ is referred to as "elemental Fisher Information Matrix" (eFIM).

Let us consider model (4), in which case $\boldsymbol{\zeta} = (\mu, \sigma_W^2, \sigma_\mu^2)^\top$. From (5) it follows that

$$\frac{\partial \boldsymbol{\mu}(\boldsymbol{\zeta})}{\partial \mu} = \boldsymbol{\tau}, \quad \frac{\partial \boldsymbol{\mu}(\boldsymbol{\zeta})}{\partial \sigma_W^2} = \frac{\partial \boldsymbol{\mu}(\boldsymbol{\zeta})}{\partial \sigma_\mu^2} = \mathbf{0}, \tag{7}$$

$$\frac{\partial \boldsymbol{\Sigma}(\boldsymbol{\zeta})}{\partial \mu} = \mathbf{0}, \quad \frac{\partial \boldsymbol{\Sigma}(\boldsymbol{\zeta})}{\partial \sigma_W^2} = \mathbf{S}, \quad \frac{\partial \boldsymbol{\Sigma}(\boldsymbol{\zeta})}{\partial \sigma_\mu^2} = \boldsymbol{\tau}\boldsymbol{\tau}^\top, \tag{8}$$

and we have

$$\boldsymbol{\Sigma}^{-1}(\boldsymbol{\zeta}) = \frac{1}{\sigma_W^2}\left(\mathbf{S}^{-1} - \frac{\sigma_\mu^2 \boldsymbol{\ell}\boldsymbol{\ell}^\top}{\sigma_W^2 + \sigma_\mu^2 \mathcal{T}}\right), \tag{9}$$

where the above formula follows from [18], Theorem 18.2.8; see also [16], Eq. (10.13). In (9), $\boldsymbol{\ell} = (1, \ldots, 1)^\top$ ($\dim \boldsymbol{\ell} = n$), and $\mathcal{T} = \boldsymbol{\tau}^\top \boldsymbol{\ell} = t_n - t_0$ is the length of the patient's observation period ("exposure").

Ultimately, for model (4), eFIM for $\boldsymbol{\zeta} = (\mu, \sigma_W^2, \sigma_\mu^2)^\top$ can be expressed as

$$\mathscr{M} = \begin{pmatrix} \omega & 0 & 0 \\ 0 & \frac{n-1}{2\sigma_W^4} + \frac{\omega^2}{2\mathcal{T}^2} & \frac{\omega^2}{2\mathcal{T}} \\ 0 & \frac{\omega^2}{2\mathcal{T}} & \frac{\omega^2}{2} \end{pmatrix}, \quad \omega = \frac{\mathcal{T}}{\sigma_W^2 + \sigma_\mu^2 \mathcal{T}}, \tag{10}$$

and the inverse of $\mathscr{M}$ is

$$\mathbf{D} = \mathscr{M}^{-1} = \begin{pmatrix} \frac{1}{\omega} & 0 & 0 \\ 0 & \frac{2\sigma_W^4}{n-1} & -\frac{2\sigma_W^4}{(n-1)\mathcal{T}} \\ 0 & -\frac{2\sigma_W^4}{(n-1)\mathcal{T}} & \frac{2}{\omega^2} + \frac{2\sigma_W^4}{(n-1)\mathcal{T}^2} \end{pmatrix}; \tag{11}$$

see Appendix A for the derivations.

In the case when population variability can be neglected, i.e., $\mu_i = \mu$, the dimension of the parameter vector $\boldsymbol{\zeta}$ is reduced to 2, i.e., $\boldsymbol{\zeta} = (\mu, \sigma_W^2)^\top$, and the eFIM (10) and its inverse simplify to

$$\tilde{\mathscr{M}} = \begin{pmatrix} \frac{\mathcal{T}}{\sigma_W^2} & 0 \\ 0 & \frac{n}{2\sigma_W^4} \end{pmatrix} \quad \text{and} \quad \tilde{\mathbf{D}} = \tilde{\mathscr{M}}^{-1} = \begin{pmatrix} \frac{\sigma_W^2}{\mathcal{T}} & 0 \\ 0 & \frac{2\sigma_W^4}{n} \end{pmatrix}. \tag{12}$$

Remark 1 The eFIM $\mathscr{M}$ in (10) does not depend on the allocation of observations on the time scale (the values $\{\tau_j\}_1^n$), but does depend on the length of the observation period $\mathcal{T}$ and the number of measurements n. In other words, two subjects are informationally equivalent if their vectors $\boldsymbol{\tau}$ yield the same values of $\mathcal{T}$ and n. However, in general, measurement costs may depend on the exact timing of the measurements. Throughout the paper, for the sake of simplicity, we assume that the cost is entirely defined by $\mathcal{T}$ and n.

Remark 2 Matrices $\mathbf{D}$ and $\tilde{\mathbf{D}}$ are non-increasing functions (in the Loewner ordering sense) of $\mathcal{T}$ and n. While increasing these parameters can statistically reduce the required number of study participants (N), practical constraints must be considered: excessive observation period ($\mathcal{T}$) may exceed the model's validity range, while high sampling frequency (n) may be medically irrelevant or operationally infeasible in a clinical setting.

3.2 FIM of a single patient in the presence of covariates

Suppose μ, σ_W^2, and σ_μ^2 depend on some covariates, such as age, sex, genetic signature, drug dose, etc., and this dependence can be parameterized, i.e., $\zeta_1 = \mu = \mu(\mathbf{x}, \boldsymbol{\theta})$, $\zeta_2 = \sigma_W^2 = \sigma_W^2(\mathbf{x}, \boldsymbol{\theta})$, and $\zeta_3 = \sigma_\mu^2 = \sigma_\mu^2(\mathbf{x}, \boldsymbol{\theta})$, where $\mathbf{x}$ is the vector of covariates and $\boldsymbol{\theta}$ is a vector of parameters to be estimated. In other words, we consider a "regression model" setting. The eFIM for $\boldsymbol{\theta}$ can be found as ([16], Ch. 1.6):

$$\mathbf{M}(\mathbf{x}|\boldsymbol{\theta}, \mathcal{T}, n) = \mathbf{F}(\mathbf{x}, \boldsymbol{\theta}) \mathscr{M}(\boldsymbol{\zeta}) \mathbf{F}^\top(\mathbf{x}, \boldsymbol{\theta}), \tag{13}$$

where $\mathscr{M}(\boldsymbol{\zeta})$ is given by (10), and $\mathbf{F}(\mathbf{x}, \boldsymbol{\theta})$ is the Jacobian matrix of order $m \times m'$:

$$\mathbf{F}(\mathbf{x}, \boldsymbol{\theta}) = \frac{\partial \boldsymbol{\zeta}^\top}{\partial \boldsymbol{\theta}} = \begin{pmatrix} \frac{\partial \zeta_1}{\partial \theta_1} & \cdots & \frac{\partial \zeta_{m'}}{\partial \theta_1} \\ \cdots & \cdots & \cdots \\ \frac{\partial \zeta_1}{\partial \theta_m} & \cdots & \frac{\partial \zeta_{m'}}{\partial \theta_m} \end{pmatrix}, \quad m' = \dim \boldsymbol{\zeta}, \quad m = \dim \boldsymbol{\theta}. \tag{14}$$

In the case when $\boldsymbol{\zeta} = (\mu, \sigma_W^2, \sigma_\mu^2)^\top$, (14) is the $m \times 3$ matrix

$$\mathbf{F}(\mathbf{x}, \boldsymbol{\theta}) = \left(\frac{\partial \mu(\mathbf{x}, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}}, \frac{\partial \sigma_W^2(\mathbf{x}, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}}, \frac{\partial \sigma_\mu^2(\mathbf{x}, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \right), \tag{15}$$

and substituting it in (13), one can verify that

$$\begin{aligned} \mathbf{M}(\mathbf{x}|\boldsymbol{\theta}, \mathcal{T}, n) &= \omega \frac{\partial \mu}{\partial \boldsymbol{\theta}} \frac{\partial \mu}{\partial \boldsymbol{\theta}^\top} + \frac{n-1}{2\sigma_W^4} \frac{\partial \sigma_W^2}{\partial \boldsymbol{\theta}} \frac{\partial \sigma_W^2}{\partial \boldsymbol{\theta}^\top} \\ &\quad + \frac{\omega^2}{2} \left(\frac{1}{\mathcal{T}} \frac{\partial \sigma_W^2}{\partial \boldsymbol{\theta}} + \frac{\partial \sigma_\mu^2}{\partial \boldsymbol{\theta}} \right) \left(\frac{1}{\mathcal{T}} \frac{\partial \sigma_W^2}{\partial \boldsymbol{\theta}} + \frac{\partial \sigma_\mu^2}{\partial \boldsymbol{\theta}} \right)^\top, \end{aligned} \tag{16}$$

where ω is introduced in (10), and the dependence of the right-hand side of (16) on $\mathbf{x}$ is omitted for simplicity of notation.

In the case when the population variability can be neglected, we have $\boldsymbol{\zeta} = (\mu, \sigma_W^2)^\top$, and (13) takes the form

$$\begin{aligned} \mathbf{M}(\mathbf{x}|\boldsymbol{\theta}, \mathcal{T}, n) &= \frac{\mathcal{T}}{\sigma_W^2} \frac{\partial \mu}{\partial \boldsymbol{\theta}} \frac{\partial \mu}{\partial \boldsymbol{\theta}^\top} + \frac{n}{2\sigma_W^4} \frac{\partial \sigma_W^2}{\partial \boldsymbol{\theta}} \frac{\partial \sigma_W^2}{\partial \boldsymbol{\theta}^\top} \\ &= \frac{\mathcal{T}}{\sigma_W^2} \frac{\partial \mu}{\partial \boldsymbol{\theta}} \frac{\partial \mu}{\partial \boldsymbol{\theta}^\top} + \frac{n}{2} \frac{\partial \ln \sigma_W^2}{\partial \boldsymbol{\theta}} \frac{\partial \ln \sigma_W^2}{\partial \boldsymbol{\theta}^\top}. \end{aligned} \tag{17}$$

In (17), the logarithmic presentation of the term $\frac{\partial \ln \sigma_W^2}{\partial \boldsymbol{\theta}}$ is particularly advantageous because it simplifies the derivation of the FIM when the within-process variance is modeled as an exponential function of covariates (cf. Example 1 in §4.1).

3.3 Optimal design problems with identical DAPs

Let us consider the following scheme of a clinical study:

- Let $\mathfrak{X} \in \mathbb{R}^p$ denote the design region, assumed to be a compact set. Denote $\{\mathbf{x}_k\}_1^K \subset \mathfrak{X}$ the design points. Each patient enrolled in the study is assigned (at random) to one of $\mathbf{x}_k$'s. Thus, we assume all covariates can be controlled.
- N_k patients are assigned to the respective $\mathbf{x}_k$. In general, n_{ik} measurements can be performed on the ith patient during $\mathcal{T}_{ik}$ time interval. Thus, at every $\mathbf{x}_k$, a total of $\mathscr{N}_k = \sum_{i=1}^{N_k} n_{ik}$ measurements are performed during gross time interval $\mathscr{T}_k = \sum_{i=1}^{N_k} \mathcal{T}_{ik}$. However, in a randomized controlled trial (RCT), it is desirable to keep $n_{ik} \equiv n$ and $\mathcal{T}_{ik} \equiv \mathcal{T}$, to assure the double-masked nature of the study. The latter requirement (identical DAP for all participants in the study) will be assumed for the rest of this subsection.

The collection

$$\underline{\xi}_N = \{\mathbf{x}_k, N_k | \mathcal{T}, n\}_1^K \tag{18}$$

is referred to as *study design*. The FIM for $\boldsymbol{\theta}$ associated with (18) is

$$\underline{\mathbf{M}}(\underline{\xi}_N|\boldsymbol{\theta},\mathcal{T},n)=\sum_{k=1}^{K}N_k\mathbf{M}(\mathbf{x}_k|\boldsymbol{\theta},\mathcal{T},n). \tag{19}$$

Introducing the "normalized" study design

$$\xi_N=\{\mathbf{x}_k,p_k|\mathcal{T},n\}_1^K,\quad p_k=N_k/N, \tag{20}$$

and the "normalized" FIM

$$\mathbf{M}(\xi_N|\boldsymbol{\theta},\mathcal{T},n)=\sum_{k=1}^{K}p_k\mathbf{M}(\mathbf{x}_k|\boldsymbol{\theta},\mathcal{T},n), \tag{21}$$

and resorting to the continuous versions of (20) and (21) when the discreteness of p_k is neglected and, moreover, design ξ can be viewed as the probability measure defined on $\mathfrak{X}$, we can formulate the following optimization problem:

$$\xi^*(\boldsymbol{\theta}|\mathcal{T},n)=\arg\min_{\xi}\Psi[\mathbf{M}(\xi|\boldsymbol{\theta},\mathcal{T},n)], \tag{22}$$

where

$$\mathbf{M}(\xi|\boldsymbol{\theta},\mathcal{T},n)=\int_{\mathbf{x}\in\mathfrak{X}}\mathbf{M}(\mathbf{x}|\boldsymbol{\theta},\mathcal{T},n)\xi(d\mathbf{x}), \tag{23}$$

and Ψ is a criterion satisfying the following properties ([16], Ch. 2.3):

(a) Monotonicity, i.e.,

$$\Psi[\mathbf{M}]\le\Psi[\mathbf{M}'],\quad \text{if } \mathbf{M}\ge\mathbf{M}',$$

where $\mathbf{M}$ and $\mathbf{M}'$ are nonnegatively defined matrices, and the second inequality is understood in terms of the Loewner ordering.

(b) Homogeneity, i.e.,

$$\Psi[N\mathbf{M}]=\phi(N)\Psi[\mathbf{M}],$$

where $\phi(N)$ is nonincreasing.

(c) Convexity, i.e.,

$$\Psi[(1-\alpha)\mathbf{M}+\alpha\mathbf{M}']\le(1-\alpha)\Psi[\mathbf{M}]+\alpha\Psi[\mathbf{M}'].$$

Additivity of $\mathbf{M}(\xi|\boldsymbol{\theta},\mathcal{T},n)$ and assumptions (a)–(c) lead to the necessary and sufficient conditions that coincide with those of the General Equivalence Theorem in the traditional design of experiments (DOE) paradigm ([16], Ch. 2.4.2, Theorem 2.2, and Tables 2.1 & 5.1).

Unlike the "traditional" design, (22) is optimal given $\mathcal{T}$ and n. In general, we may extend the optimization with the proper selection of $\mathcal{T}$ and n:

$$(\mathcal{T}^*,n^*)\ =\ \arg\min_{(\mathcal{T},n)}\Psi[\mathbf{M}(\xi^*|\boldsymbol{\theta},\mathcal{T},n)] \tag{24}$$

$$\text{s.t. } \Phi(\mathcal{T}, n) \leq \Phi^* \tag{25}$$

where Φ is the cost/risk function comprising operational, ethical, and financial constraints. Following Remark 2, one can verify that (25) can be replaced with the equality $\Phi(\mathcal{T}, n) = \Phi^*$, and the optimization problem (24)–(25) is actually the univariate one. Importantly, it involves only $\mathcal{T} = \boldsymbol{\ell}^\top \boldsymbol{\tau}$ and $n = \dim \boldsymbol{\tau}$, but not the components of $\boldsymbol{\tau}$.

In Section 4, we consider several special cases of optimization problem (22), highlighting the specifics which make it different from the traditional DOE.

4 Special cases and examples

4.1 The case when population variability can be neglected

Suppose $\sigma_\mu^2(\mathbf{x}, \boldsymbol{\theta}) \equiv 0$, in which case eFIM for $\boldsymbol{\theta}$ is given by (17) with $\mu = \mu(\mathbf{x}, \boldsymbol{\theta})$ and $\sigma_W^2 = \sigma_W^2(\mathbf{x}, \boldsymbol{\theta})$. The respective "normalized" FIM for design ξ is

$$\mathbf{M}(\xi|\boldsymbol{\theta}, \mathcal{T}, n) = \sum_{k=1}^{K} p_k \mathbf{M}(\mathbf{x}_k|\boldsymbol{\theta}, \mathcal{T}, n) = \mathcal{T} \sum_{k=1}^{K} p_K \tilde{\mathbf{M}}(\mathbf{x}_k|\boldsymbol{\theta}, \Delta), \tag{26}$$

where

$$\tilde{\mathbf{M}}(\mathbf{x}_k|\boldsymbol{\theta}, \Delta) = \frac{1}{\sigma_W^2} \frac{\partial \mu}{\partial \boldsymbol{\theta}} \frac{\partial \mu}{\partial \boldsymbol{\theta}^\top} + \frac{1}{2\Delta} \frac{\partial \ln \sigma_W^2}{\partial \boldsymbol{\theta}} \frac{\partial \ln \sigma_W^2}{\partial \boldsymbol{\theta}^\top}, \quad \Delta = \mathcal{T}/n. \tag{27}$$

Remark 3 In clinical studies, Δ is often defined by circadian rhythms, such as day, week, month, etc. In general, Δ can be viewed as an average time interval between the neighboring observations.

We can now formulate the following "continuous" optimal design problem:

Step 1: Find

$$\xi^*(\boldsymbol{\theta}, \Delta) = \arg\min_{\xi} \Psi[\tilde{\mathbf{M}}(\xi|\boldsymbol{\theta}, \Delta)], \tag{28}$$

where

$$\tilde{\mathbf{M}}(\xi|\boldsymbol{\theta}, \Delta) = \int_{\mathbf{x} \in \mathfrak{X}} \tilde{\mathbf{M}}(\mathbf{x}|\boldsymbol{\theta}, \Delta) \xi(d\mathbf{x}).$$

Step 2: Select the precision level Ψ^* and determine Q^* such that

$$\Psi^* = \phi(Q^*) \Psi[\tilde{\mathbf{M}}(\xi^*|\boldsymbol{\theta}, \Delta)], \tag{29}$$

and minimize the cost $\Phi(\mathcal{T}, N)$ given cumulative exposure $\mathcal{T} N = Q^*$. Some additional "common" constraints may have to be imposed on $\mathcal{T}$ and N; for instance, $\mathcal{T}$ should be within the model validity range. The outcome of the two-step procedure contains the "continuous" optimal design ξ^* and the optimal combination of $\mathcal{T}^*$ and N^*.

In what follows, the "necessary and sufficient conditions" for optimality of a design $\xi = \xi(\boldsymbol{\theta}, \Delta)$ will be used. These conditions are a part of the General Equivalence

Theorem; see, for instance, [16], Ch. 2.4.2, Theorem 2.1 and Table 2.1. We will use shorter notations, $\tilde{\mathbf{D}}(\xi) = \tilde{\mathbf{M}}^{-1}(\xi|\boldsymbol{\theta}, \Delta)$ for the dispersion matrix of the design ξ, and $\tilde{\mathbf{M}}(\mathbf{x}) = \tilde{\mathbf{M}}(\mathbf{x}|\boldsymbol{\theta}, \Delta)$ for eFIM of $\boldsymbol{\theta}$ at the design point $\mathbf{x} \in \mathfrak{X}$; see (27).

For the D-criterion $\Psi(\xi) = \ln\det \tilde{\mathbf{D}}(\xi)$, the necessary and sufficient conditions are

$$\mathrm{tr}[\tilde{\mathbf{M}}(\mathbf{x})\tilde{\mathbf{D}}(\xi)] \leq \dim \boldsymbol{\theta}, \tag{30}$$

and for the linear criterion $\Psi(\xi) = \mathrm{tr}[\mathbf{A}\tilde{\mathbf{D}}(\xi)]$, they are

$$\mathrm{tr}[\tilde{\mathbf{M}}(\mathbf{x})\tilde{\mathbf{D}}(\xi)\mathbf{A}\tilde{\mathbf{D}}(\xi)] \leq \mathrm{tr}[\mathbf{A}\tilde{\mathbf{D}}(\xi)], \tag{31}$$

for all $\mathbf{x} \in \mathfrak{X}$. In (31), $\mathbf{A}$ is frequently referred to as "utility matrix."

Example 1: Two-arm RCT

Consider an RCT comparing the effects of two treatments on disease progression. Following subsection 3.2 with the simplest setting of one covariate and two linear functions, we have

$$\mu(x, \boldsymbol{\theta}) = \theta_1 + \theta_2 x \quad \text{and} \quad \ln \sigma_W^2(x, \boldsymbol{\theta}) = \theta_3 + \theta_4 x,$$

where $x = -1$ corresponds to the control (C) and $x = 1$ corresponds to the experimental treatment (E). Introducing $\boldsymbol{\theta} = (\theta_1, \theta_2, \theta_3, \theta_4)^\top$, we have

$$\frac{\partial \mu(x, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = (1, x, 0, 0)^\top, \quad \text{and} \quad \frac{\partial \ln \sigma_W^2(x, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = (0, 0, 1, x)^\top.$$

The single-patient FIM is a 4×4 block-diagonal matrix

$$\begin{aligned} \mathbf{M}(x|\boldsymbol{\theta}, \mathcal{T}, n) &= \mathcal{T} \begin{pmatrix} \frac{1}{\sigma_W^2(x,\boldsymbol{\theta})}\mathbf{f}(x)\mathbf{f}^\top(x) & \mathbf{0} \\ \mathbf{0} & \frac{1}{2\Delta}\mathbf{f}(x)\mathbf{f}^\top(x) \end{pmatrix} \\ &= \mathcal{T}\tilde{\mathbf{M}}(x|\boldsymbol{\theta}, \Delta), \end{aligned} \tag{32}$$

where $\mathbf{f}(x) = (1, x)^\top$. For notation convenience, define group-specific parameters:

$$\mu_C = \mu(-1, \boldsymbol{\theta}),\ \sigma_C^2 = \sigma_W^2(-1, \boldsymbol{\theta}) \quad \text{and} \quad \mu_E = \mu(1, \boldsymbol{\theta}),\ \sigma_E^2 = \sigma_W^2(1, \boldsymbol{\theta}).$$

In the considered setting, all possible normalized designs can be described as

$$\xi = \xi(p) = \left\{ \begin{matrix} -1 & 1 \\ 1-p & p \end{matrix} \right\}, \quad 0 \leq p \leq 1.$$

Given $\boldsymbol{\theta}$ and Δ, the FIM of $\xi(p)$ can be written as

$$\tilde{\mathbf{M}}(p|\boldsymbol{\theta}, \Delta) = \mathcal{T} \begin{pmatrix} \tilde{\mathbf{M}}_{11}(p|\boldsymbol{\theta}, \Delta) & \mathbf{0} \\ \mathbf{0} & \tilde{\mathbf{M}}_{22}(p|\boldsymbol{\theta}, \Delta) \end{pmatrix}, \tag{33}$$

where

$$\tilde{\mathbf{M}}_{11}(p|\boldsymbol{\theta},\Delta) = \frac{1-p}{\sigma_C^2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} + \frac{p}{\sigma_E^2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} \frac{p}{\sigma_E^2} + \frac{1-p}{\sigma_C^2} & \frac{p}{\sigma_E^2} - \frac{1-p}{\sigma_C^2} \\ \frac{p}{\sigma_E^2} - \frac{1-p}{\sigma_C^2} & \frac{p}{\sigma_E^2} + \frac{1-p}{\sigma_C^2} \end{pmatrix}, \tag{34}$$

and

$$\tilde{\mathbf{M}}_{22}(p|\boldsymbol{\theta},\Delta) = \frac{1}{2\Delta}\begin{pmatrix} 1 & 2p-1 \\ 2p-1 & 1 \end{pmatrix}. \tag{35}$$

The inverse of (33) is a 4×4 matrix with two diagonal blocks

$$\tilde{\mathbf{M}}_{11}^{-1}(p|\boldsymbol{\theta},\Delta) = \begin{pmatrix} \frac{\sigma_E^2}{p} + \frac{\sigma_C^2}{1-p} & \frac{\sigma_E^2}{p} - \frac{\sigma_C^2}{1-p} \\ \frac{\sigma_E^2}{p} - \frac{\sigma_C^2}{1-p} & \frac{\sigma_E^2}{p} + \frac{\sigma_C^2}{1-p} \end{pmatrix}, \tag{36}$$

and

$$\tilde{\mathbf{M}}_{22}^{-1}(p|\boldsymbol{\theta},\Delta) = \frac{\Delta}{2p(1-p)}\begin{pmatrix} 1 & 1-2p \\ 1-2p & 1 \end{pmatrix}. \tag{37}$$

Suppose the nominal values of (θ_3,θ_4) are available, i.e., $\sigma_C^2 = \exp(\theta_3-\theta_4)$ and $\sigma_E^2 = \exp(\theta_3+\theta_4)$ are known. In this case, the normalized variance-covariance matrix is $\tilde{\mathbf{D}}(p) = \tilde{\mathbf{M}}_{11}^{-1}(p|\boldsymbol{\theta},\Delta)$, and the determinant $\det\tilde{\mathbf{D}}(p) = \frac{4\sigma_E^2\sigma_C^2}{p(1-p)}$ is minimized for $p^* = 0.5$ (equal allocation). If the primary goal is to estimate the treatment effect $\mu_E-\mu_C = 2\theta_2$, then the normalized variance of its efficient estimator (see (36)) is

$$(N\mathcal{T})^{-1}\mathrm{Var}[2\hat{\theta}_2] = 4\left(\frac{\sigma_E^2}{p} + \frac{\sigma_C^2}{1-p}\right),$$

which is minimized for

$$p^* = \frac{\sigma_E}{\sigma_E+\sigma_C}, \tag{38}$$

also known as the Neyman allocation [19].

Suppose all four components of $\boldsymbol{\theta} = (\theta_1,\theta_2,\theta_3,\theta_4)^\top$ are of interest. Then

$$\det\tilde{\mathbf{D}}(p) = \det\tilde{\mathbf{M}}_{11}^{-1}(p|\boldsymbol{\theta},\Delta)\cdot\det\tilde{\mathbf{M}}_{22}^{-1}(p|\boldsymbol{\theta},\Delta) = \frac{\Delta^2\sigma_E^2\sigma_C^2}{p^2(1-p)^2},$$

is minimized for $p^* = 0.5$.

If one is interested in the joint estimation of $\mu_E-\mu_C = 2\theta_2$ and $\ln(\sigma_E^2/\sigma_C^2) = 2\theta_4$, then the utility matrix is (see (31)) is

$$\mathbf{A} = \mathbf{L}\mathbf{L}^\top, \quad \mathbf{L}^\top = \begin{pmatrix} 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}.$$

The corresponding A-criterion is written as

$$\mathrm{tr}[\mathbf{A}\tilde{\mathbf{D}}(p)] = 4\left(\frac{\sigma_E^2}{p} + \frac{\sigma_C^2}{1-p} + \frac{\Delta^2}{p(1-p)}\right),$$

which is minimized for

$$p^* = \frac{\sqrt{\sigma_E^2 + \Delta^2}}{\sqrt{\sigma_E^2 + \Delta^2} + \sqrt{\sigma_C^2 + \Delta^2}}. \tag{39}$$

Notice that (39) tends to Neyman allocation (38), if $\Delta \to 0$; and it tends to 0.5, if $\Delta \to \infty$.

Example 2: S-shape dose–response curve

Let

$$\mu = \frac{e^\gamma}{1+e^\gamma}, \quad \gamma = \frac{x-\theta_1}{\theta_2}, \quad \sigma_W^2 \equiv \theta_3, \quad \text{and} \quad \mathfrak{X} = \{-\infty < x - \theta_1 < \infty\}. \tag{40}$$

Obviously,

$$\gamma = \ln \frac{\mu}{1-\mu} \quad \text{and} \quad \mu(-\gamma) = 1 - \mu(\gamma). \tag{41}$$

Following subsection 3.1, the eFIM for model (40) is

$$\tilde{\mathscr{M}}(\boldsymbol{\zeta}) = \begin{pmatrix} \mathcal{T}\sigma_W^{-2} & 0 \\ 0 & \frac{n}{2}\sigma_W^{-4} \end{pmatrix}, \quad \boldsymbol{\zeta} = (\mu, \sigma_W^2)^\top. \tag{42}$$

Observing that

$$\frac{\partial \mu}{\partial \theta_1} = -\frac{\mu(1-\mu)}{\theta_2}, \quad \frac{\partial \mu}{\partial \theta_2} = -\frac{\gamma\mu(1-\mu)}{\theta_2}, \quad \frac{\partial \sigma_W^2}{\partial \theta_3} = 1,$$

and

$$\mathbf{F}^\top = \begin{pmatrix} \frac{\partial \mu}{\partial \theta_1} & \frac{\partial \mu}{\partial \theta_2} & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

one can verify (see (17)) that the single-patient FIM in this case is

$$\mathbf{M}(x|\boldsymbol{\theta}, \Delta) = \mathcal{T}\begin{pmatrix} \mathbf{M}_{11} & \mathbf{0} \\ \mathbf{0} & \frac{2}{\Delta}\theta_3 \end{pmatrix}, \quad \mathbf{M}_{11} = \frac{\mu^2(1-\mu)^2}{\theta_2^2}\begin{pmatrix} 1 & \ln\frac{\mu}{1-\mu} \\ \ln\frac{\mu}{1-\mu} & \ln^2\frac{\mu}{1-\mu} \end{pmatrix}. \tag{43}$$

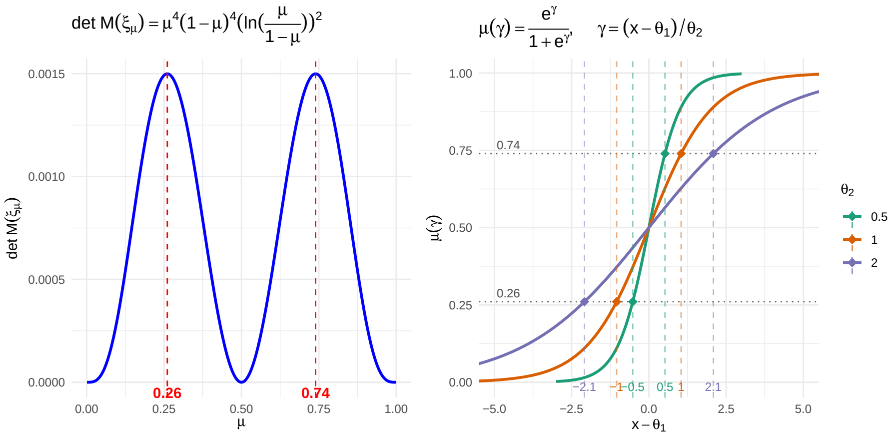
The idea of mapping from x-space to μ-space was intensively explored in [20] in the "traditional" DOE setting. In the present setting, however, the FIM is proportional to $N\mathcal{T}$, whereas in the traditional setting it scales only with N.

Identities (41) suggest that designs

$$\xi = \left\{ \begin{matrix} x_1 & \dots & x_K \\ p_1 & \dots & p_K \end{matrix} \right\} \quad \text{and} \quad \xi' = \left\{ \begin{matrix} 2\theta_1 - x_1 & \dots & 2\theta_1 - x_K \\ p_1 & \dots & p_K \end{matrix} \right\} \tag{44}$$

have exactly the same FIM. The same statement is true for ξ_μ, the respective design measures in the μ-space. Therefore, it is expedient to consider two-point designs of

Fig. 1 D-optimal design points in the μ-space (left plot) and in the x-space (right plot).



the form

$$\xi_\mu = \begin{Bmatrix} \mu & 1-\mu \\ 0.5 & 0.5 \end{Bmatrix}.$$

For these designs,

$$\mathbf{M}(\xi_\mu|\boldsymbol{\theta},\Delta) = \mathcal{T}\begin{pmatrix} \tilde{\mathbf{M}}_{11} & \mathbf{0} \\ \mathbf{0} & \frac{2}{\Delta}\theta_3 \end{pmatrix}, \quad \tilde{\mathbf{M}}_{11} = \frac{\mu^2(1-\mu)^2}{\theta_2^2}\begin{pmatrix} 1 & 0 \\ 0 & \ln^2\frac{\mu}{1-\mu} \end{pmatrix}, \tag{45}$$

and

$$\det \mathbf{M}(\xi_\mu|\boldsymbol{\theta},\Delta) \propto \mu^4(1-\mu)^4 \ln^2\frac{\mu}{1-\mu}. \tag{46}$$

Figure 1 provides a graphical solution to the D-optimal design problem. Interestingly, in the μ-space, the two support points are 0.26 and 0.74. In the x-space, the location of the optimal support points is sensitive to the values of the parameters which define the dose–response curve.

4.2 The case when population variability cannot be neglected

We examine two modeling frameworks for scenarios where between-patient variability is a significant factor. In Type I models, a generalized regression structure is applied directly to the mean drift, the Wiener process variance, and the population variance, following the methodology described in subsection 3.2. In contrast, Type II models utilize regression to characterize the drift and within-process variance at the individual patient level, while further assuming that the parameters governing each patient's drift are randomly sampled from a normal population distribution.

For a Type I model, let us assume

$$\bar{\mu} = \boldsymbol{\theta}_1^\top \mathbf{f}_1(\mathbf{x}), \quad \sigma_W^2 = \exp(\boldsymbol{\theta}_2^\top \mathbf{f}_2(\mathbf{x})), \quad \sigma_\mu^2 = \exp(\boldsymbol{\theta}_3^\top \mathbf{f}_3(\mathbf{x})), \tag{47}$$

where $\dim \boldsymbol{\theta}_k = m_k$, $k = 1, 2, 3$, $m = m_1 + m_2 + m_3$, and $\boldsymbol{\theta}^\top = (\boldsymbol{\theta}_1^\top, \boldsymbol{\theta}_2^\top, \boldsymbol{\theta}_3^\top)$. Using (16), one can verify that the single-patient FIM for $\boldsymbol{\theta}$ is

$$\mathbf{M}(\mathbf{x}|\boldsymbol{\theta}, \mathcal{T}, n) = \begin{pmatrix} \omega \mathbf{f}_1 \mathbf{f}_1^\top & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \left(\frac{n-1}{2\sigma_W^4} + \frac{\omega^2}{2\mathcal{T}^2}\right) \sigma_W^4 \mathbf{f}_2 \mathbf{f}_2^\top & \frac{\omega^2}{2\mathcal{T}} \sigma_W^2 \sigma_\mu^2 \mathbf{f}_2 \mathbf{f}_3^\top \\ \mathbf{0} & \frac{\omega^2}{2\mathcal{T}} \sigma_W^2 \sigma_\mu^2 \mathbf{f}_3 \mathbf{f}_2^\top & \frac{\omega^2}{2} \sigma_\mu^4 \mathbf{f}_3 \mathbf{f}_3^\top \end{pmatrix}. \tag{48}$$

For simplicity of notation, the dependence of the right-hand side of (48) on $\mathbf{x}$ and $\boldsymbol{\theta}$ is omitted.

For a Type II model, let us assume that $\mu_i = \boldsymbol{\theta}_{1i}^\top \mathbf{f}_1(\mathbf{x})$ and $\boldsymbol{\theta}_{1i} \sim \mathcal{N}(\bar{\boldsymbol{\theta}}_1, \boldsymbol{\Sigma})$. Then, instead of (47), we have

$$\bar{\mu} = \bar{\boldsymbol{\theta}}_1^\top \mathbf{f}_1(\mathbf{x}), \quad \sigma_W^2 = \exp(\boldsymbol{\theta}_2^\top \mathbf{f}_2(\mathbf{x})), \quad \sigma_\mu^2 = \mathbf{f}_1^\top(\mathbf{x}) \boldsymbol{\Sigma} \mathbf{f}_1(\mathbf{x}), \tag{49}$$

where

$$\bar{\boldsymbol{\theta}}_1 = \mathrm{E}[\boldsymbol{\theta}_1] \quad \text{and} \quad \boldsymbol{\Sigma} = \mathrm{Var}[\boldsymbol{\theta}_1],$$

with $m_1 = \dim \bar{\boldsymbol{\theta}}_1$, $m_2 = \dim \boldsymbol{\theta}_2$, $m_3 = m_1(m_1+1)/2$. The respective FIM, i.e., analogue of (48), may be of large dimension. In the simplest case of the diagonal matrix $\boldsymbol{\Sigma} = \mathrm{diag}\{\Sigma_{\alpha\alpha}\}_1^{m_1}$, we have

$$\mathbf{M}(\mathbf{x}|\boldsymbol{\theta}, \mathcal{T}, n) = \begin{pmatrix} \omega \mathbf{f}_1 \mathbf{f}_1^\top & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \left(\frac{n-1}{2\sigma_W^4} + \frac{\omega^2}{2\mathcal{T}^2}\right) \sigma_W^4 \mathbf{f}_2 \mathbf{f}_2^\top & \frac{\omega^2}{2\mathcal{T}} \sigma_W^2 \mathbf{f}_2 \boldsymbol{\varphi}^\top \\ \mathbf{0} & \frac{\omega^2}{2\mathcal{T}} \sigma_W^2 \mathbf{f}_2 \boldsymbol{\varphi}^\top & \frac{\omega^2}{2} \boldsymbol{\varphi} \boldsymbol{\varphi}^\top \end{pmatrix}, \tag{50}$$

where $\boldsymbol{\varphi} = \{\varphi_\alpha(\mathbf{x})\}_1^{m_1}$, $\varphi_\alpha(\mathbf{x}) = f_{1\alpha}^2(\mathbf{x})$, and $\boldsymbol{\theta}_3 = \{\Sigma_{\alpha\alpha}\}_1^{m_1}$, $\alpha = 1, \ldots, m_1$.

A major problem with Type II case is potentially large m_3, even for moderate m_1. Also, if $\mu_i = \eta(\mathbf{x}, \boldsymbol{\theta}_{1i})$ is a nonlinear function of $\eta(\cdot)$, then

$$\bar{\mu} \simeq \eta(\mathbf{x}, \bar{\boldsymbol{\theta}}_1) + \frac{1}{2} \mathrm{tr} \left[\boldsymbol{\Sigma} \frac{\partial^2 \eta(\mathbf{x}, \boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}^\top} \Big|_{\boldsymbol{\theta} = \bar{\boldsymbol{\theta}}_1} \right] \quad \text{and} \quad \sigma_\mu^2 \simeq \frac{\partial \eta(\mathbf{x}, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}^\top} \boldsymbol{\Sigma} \frac{\partial \eta(\mathbf{x}, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \Big|_{\boldsymbol{\theta} = \bar{\boldsymbol{\theta}}_1}.$$

Remark 4 Unlike in subsection 4.1, matrices (48) and (50) cannot be presented as the product of $\mathcal{T}$ and some matrix $\tilde{\mathbf{M}}(\mathbf{x}|\boldsymbol{\theta}, \Delta)$ which depends only on $\boldsymbol{\theta}$ and $\Delta = \mathcal{T}/n$. Therefore, in what follows, optimal designs will generally depend on $\boldsymbol{\theta}$, $\mathcal{T}$, and n.

Example 3: Two-arm RCT with alternative approaches for modeling population heterogeneity

Suppose the parameters in (47) are modeled as

$$\bar{\mu} = \theta_1 + \theta_2 x, \quad \sigma_W^2 = \exp(\theta_3 + \theta_4 x), \quad \text{and} \quad \sigma_\mu^2 = \exp(\theta_5 + \theta_6 x). \tag{51}$$

Therefore, we have a Type I model with $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_6)^\top$ and $\mathbf{f}_1(\mathbf{x}) = \mathbf{f}_2(\mathbf{x}) = \mathbf{f}_3(\mathbf{x}) = (1, x)^\top$. The FIM of design $\xi = \xi(p)$, $0 \le p \le 1$ is a 6×6 matrix

$$\mathbf{M}(\xi|\boldsymbol{\theta}, \mathcal{T}, n) = (1-p)\mathbf{M}(x = -1|\boldsymbol{\theta}, \mathcal{T}, n) + p\mathbf{M}(x = 1|\boldsymbol{\theta}, \mathcal{T}, n), \tag{52}$$

with $\mathbf{M}(x|\boldsymbol{\theta}, \mathcal{T}, n)$ of the form (48). Similar to subsection 4.1, various optimal design problems can be formulated and solved. For example, if the nominal values of $(\theta_3, \theta_4, \theta_5, \theta_6)^\top$ are available or out of interest in the D-optimality case, then only the upper diagonal block of (52) corresponding to $(\theta_1, \theta_2)^\top$ is of interest:

$$\mathbf{M}_{11}(p|\boldsymbol{\theta}, \mathcal{T}, n) = (1-p)\omega(-1)\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} + p\omega(1)\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \tag{53}$$

where

$$\omega(x) = \frac{\mathcal{T}}{e^{\theta_3 + \theta_4 x} + \mathcal{T} e^{\theta_5 + \theta_6 x}}. \tag{54}$$

Using similar arguments as in subsection 4.1 Example 1, the D-optimal design for estimating $(\theta_1, \theta_2)^\top$ is $p^* = 0.5$, and the A-optimal design for estimating $2\theta_2$ is

$$p^* = \frac{1/\sqrt{\omega(1)}}{1/\sqrt{\omega(1)} + 1/\sqrt{\omega(-1)}} = \frac{\sqrt{e^{\theta_3+\theta_4} + \mathcal{T} e^{\theta_5+\theta_6}}}{\sqrt{e^{\theta_3+\theta_4} + \mathcal{T} e^{\theta_5+\theta_6}} + \sqrt{e^{\theta_3-\theta_4} + \mathcal{T} e^{\theta_5-\theta_6}}}. \tag{55}$$

Notice that if $\mathcal{T} \to \infty$, (55) tends to 0.5.

Let us consider a Type II model with $\mu_i = \boldsymbol{\theta}_{1i}^\top \mathbf{f}_1(\mathbf{x})$, $\boldsymbol{\theta}_{1i} \sim \mathcal{N}(\bar{\boldsymbol{\theta}}_1, \boldsymbol{\Sigma})$, $\bar{\boldsymbol{\theta}}_1 = (\theta_1, \theta_2)^\top$, $\mathbf{f}_1(\mathbf{x}) = (1, x)^\top$, and $\boldsymbol{\Sigma} = \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{pmatrix}$. Also, assume $\sigma_W^2 = \exp(\boldsymbol{\theta}_2^\top \mathbf{f}_2(\mathbf{x}))$, $\boldsymbol{\theta}_2 = (\theta_3, \theta_4)^\top$, and $\mathbf{f}_2(\mathbf{x}) = (1, x)^\top$. In this case, (49) is written as

$$\bar{\mu} = \theta_1 + \theta_2 x, \quad \sigma_W^2 = \exp(\theta_3 + \theta_4 x), \quad \text{and} \quad \sigma_\mu^2 = \sigma_{11} + 2\sigma_{12} x + \sigma_{22} x^2. \tag{56}$$

Suppose the nominal values of $(\theta_3, \theta_4, \sigma_{11}, \sigma_{12}, \sigma_{22})^\top$ are available, in which case the FIM for $(\theta_1, \theta_2)^\top$ of design $\xi = \xi(p)$ has the form (53) with

$$\omega(x) = \frac{\mathcal{T}}{e^{\theta_3+\theta_4 x} + \mathcal{T}(\sigma_{11} + \sigma_{12} x + \sigma_{22} x^2)}. \tag{57}$$

Using the same arguments as in the Type I case, the D-optimal design for estimating $(\theta_1, \theta_2)^\top$ is $p^* = 0.5$, and the A-optimal design for estimating $2\theta_2$ is

$$p^* = \frac{\sqrt{e^{\theta_3+\theta_4} + \mathcal{T}(\sigma_{11} + 2\sigma_{12} + \sigma_{22})}}{\sqrt{e^{\theta_3+\theta_4} + \mathcal{T}(\sigma_{11} + 2\sigma_{12} + \sigma_{22})} + \sqrt{e^{\theta_3-\theta_4} + \mathcal{T}(\sigma_{11} - 2\sigma_{12} + \sigma_{22})}}. \tag{58}$$

5 Discussion and future work

Disease progression is typically described by a stochastic process that takes into account the underlying mechanism of the phenomenon of interest and various sources of uncertainty. In the current paper, we used a Wiener process for modeling the dynamics of a disease-related biomarker in a clinical research study. A three-parameter model capturing the mean drift and variance of a Wiener process, as well as the population variance, provides a good balance between simplicity and realism. The proposed model leverages the longitudinal observations, which are assumed to follow a multivariate normal distribution, to characterize the underlying disease trajectory. This approach also enables the prediction of the probability that a patient remains below the failure threshold beyond the study duration by utilizing the properties of the inverse Gaussian distribution associated with the process's first passage time.

In our considered setting, the eFIM for the basic model parameters and its inverse are available analytically, and it is simplified even further if the population variability can be neglected. We utilized a "generalized regression" approach to model the dependence of the basic parameters—mean drift, within-process volatility, and population variance—on covariates. Our examples demonstrated the application of this methodology for optimal designs of a two-arm randomized controlled trial (RCT), focusing on treatment effects in both a linear covariate model and an S-shaped dose–response framework. We illustrated how these models capture differential treatment effects, such as the experimental therapy altering the slope of degradation and reducing the volatility of the process to stabilize disease. The formulation of an actual working model will be context-specific and require collaborative effort between the clinical investigator, statistician, and other stakeholders.

The optimal design problem formulated in Section 3 is different from that in the "classical" DOE. In our case, the experimental unit is a study patient, but associated with each patient is the "study protocol" determined by the observation period ($\mathcal{T}$) and the number of sampling time points (n). In most RCTs, to ensure a double-masked nature of the study, both $\mathcal{T}$ and n are considered to be the same across treatment groups, which is the assumption we used in our development. Ultimately, the optimization problem in Section 3 consists of two steps. For fixed $\mathcal{T}$ and n, we first find a design that is optimal under the selected criterion. We then select $(\mathcal{T}^*, n^*)$ to maximize the study criterion, subject to constraints on study duration, cost, power, etc. While the formulated problem is very general, it can be solved using available optimal design theory [16].

We showcased the utility of our proposed methodology in Section 4 by solving several optimal design problems that may arise in a two-arm RCT evaluating disease-modifying effects of an experimental therapy compared to the control, such as the standard of care. Our presented examples were in a relatively simple setting—assuming the only controllable covariate is the treatment group. More complex settings can be considered, and the presented methodology can be applied similarly. Some important considerations include the choice of a statistical model for the three basic parameters (Wiener process mean drift and variance, and the population variance). The model may be linear or nonlinear, include multiple controllable covariates and their interactions, and some covariates may be measured on a continuous scale (e.g., drug dose) while others may be chosen from a discrete set (e.g., treatment group).

It is worthwhile mentioning some additional important aspects that were outside of the scope of the current work. Our working model assumed two components of variation: within-process variance (σ_W^2), and the population variance (σ_μ^2). In some applications the measurement error cannot be neglected, in which case the covariance matrix of the vector of response increments, $\mathbf{\Sigma}_i$ in (5) includes an additional term [21], and the vector of basic parameters is $\boldsymbol{\zeta} = (\mu, \sigma_W^2, \sigma_\mu^2, \sigma_\varepsilon^2)^\top$, where σ_ε^2 is the variance of a random measurement error. The corresponding eFIM in (6) is a 4×4 matrix, and finding its inverse requires some effort. Note that σ_ε^2 is a nuisance parameter that may not be essential for the purpose of optimal study planning.

In our development, we assumed the "failure threshold" γ is sufficiently large so that the disease progression process remains below it with a probability near one. If this assumption is not feasible, the likelihood function must be adjusted to account for the specific contributions of patients who experience a threshold-crossing event. In such cases, the trajectories of "failure" patients provide different informational content than those whose observed levels remain below the threshold throughout the study. Consequently, the eFIM and the resulting optimal designs must be modified to incorporate the first-passage time distributions. We defer the formal development of these modified optimal designs to future work.

In certain applications, disease progression is characterized by several correlated processes. For instance, a bivariate model could be utilized where both components marginally follow a Wiener process structure with linear drifts, while accounting for their mutual correlation. While the derivation of the eFIM and the subsequent development of optimal designs for such multivariate systems are technically more complex, they represent a significant step toward capturing the multifaceted nature of disease trajectories. We leave the exploration of these multivariate extensions for future work.

Appendix A Proof of formulas (10) and (11)

To prove (10), let $\mathscr{M} = (\mathscr{M}_{\alpha\beta})$, where $\alpha, \beta = 1, 2, 3$. From (7), it follows immediately that $\mathscr{M}_{12} = \mathscr{M}_{13} = \mathscr{M}_{21} = \mathscr{M}_{31} = 0$. To get $\mathscr{M}_{11}$, use (5), (7), (8), and identities $\boldsymbol{\tau}^\top \boldsymbol{\ell} = \mathcal{T}$ and $\boldsymbol{\tau}^\top \mathbf{S}^{-1} \boldsymbol{\tau} = \mathcal{T}$:

$$
\begin{aligned}
\mathscr{M}_{11} &= \boldsymbol{\tau}^\top \mathbf{\Sigma}^{-1} \boldsymbol{\tau} = \frac{1}{\sigma_W^2} \left(\boldsymbol{\tau}^\top \mathbf{S}^{-1} \boldsymbol{\tau} - \frac{\sigma_\mu^2 \boldsymbol{\tau}^\top \boldsymbol{\ell}\, \boldsymbol{\ell}^\top \boldsymbol{\tau}}{\sigma_W^2 + \sigma_\mu^2 \mathcal{T}} \right) = \frac{1}{\sigma_W^2} \left(\mathcal{T} - \frac{\sigma_\mu^2 \mathcal{T}^2}{\sigma_W^2 + \sigma_\mu^2 \mathcal{T}} \right) \\
&= \frac{\mathcal{T}}{\sigma_W^2 + \sigma_\mu^2 \mathcal{T}} = \omega.
\end{aligned}
$$

For the calculation of $\mathscr{M}_{33}$, $\mathscr{M}_{23}$, and $\mathscr{M}_{22}$, use formula $\operatorname{tr}[\mathbf{AB}] = \operatorname{tr}[\mathbf{BA}]$ for $n \times k$ and $k \times n$ matrices $\mathbf{A}$ and $\mathbf{B}$, respectively. To get $\mathscr{M}_{33}$:

$$
\mathscr{M}_{33} = \frac{1}{2} \operatorname{tr}[\underbrace{\mathbf{\Sigma}^{-1} \boldsymbol{\tau}}_{\mathbf{A}} \underbrace{\boldsymbol{\tau}^\top \mathbf{\Sigma}^{-1} \boldsymbol{\tau} \boldsymbol{\tau}^\top}_{\mathbf{B}}] = \frac{1}{2} \operatorname{tr}[\underbrace{\boldsymbol{\tau}^\top \mathbf{\Sigma}^{-1} \boldsymbol{\tau} \boldsymbol{\tau}^\top}_{\mathbf{B}} \underbrace{\mathbf{\Sigma}^{-1} \boldsymbol{\tau}}_{\mathbf{A}}] = \frac{1}{2} \left(\boldsymbol{\tau}^\top \mathbf{\Sigma}^{-1} \boldsymbol{\tau} \right)^2 = \frac{\omega^2}{2}.
$$

Observing that

$$\boldsymbol{\Sigma}^{-1}\mathbf{S} = \frac{1}{\sigma_W^2}\left(\mathbf{I}_n - \frac{\sigma_\mu^2}{\sigma_W^2 + \sigma_\mu^2\mathcal{T}}\boldsymbol{\ell}\boldsymbol{\tau}^\top\right),$$

we have

$$\begin{aligned}\mathscr{M}_{23} = \mathscr{M}_{32} &= \frac{1}{2}\operatorname{tr}[\boldsymbol{\Sigma}^{-1}\boldsymbol{\tau}\boldsymbol{\tau}^\top\boldsymbol{\Sigma}^{-1}\mathbf{S}] = \frac{1}{2\sigma_W^2}\operatorname{tr}\left[\boldsymbol{\Sigma}^{-1}\boldsymbol{\tau}\boldsymbol{\tau}^\top\left(\mathbf{I}_n - \frac{\sigma_\mu^2}{\sigma_W^2 + \sigma_\mu^2\mathcal{T}}\boldsymbol{\ell}\boldsymbol{\tau}^\top\right)\right]\\ &= \frac{1}{2\sigma_W^2}\left(\omega - \omega\frac{\sigma_\mu^2\mathcal{T}}{\sigma_W^2 + \sigma_\mu^2\mathcal{T}}\right) = \frac{\omega^2}{2\mathcal{T}}.\end{aligned}$$

Similarly,

$$\begin{aligned}\mathscr{M}_{22} &= \frac{1}{2}\operatorname{tr}[\boldsymbol{\Sigma}^{-1}\mathbf{S}\boldsymbol{\Sigma}^{-1}\mathbf{S}] = \frac{1}{2}\operatorname{tr}\left[\frac{1}{\sigma_W^4}\left(\mathbf{I}_n - \frac{\sigma_\mu^2}{\sigma_W^2 + \sigma_\mu^2\mathcal{T}}\boldsymbol{\ell}\boldsymbol{\tau}^\top\right)^2\right]\\ &= \frac{1}{2\sigma_W^4}\left(n - \frac{2\sigma_\mu^2\mathcal{T}}{\sigma_W^2 + \sigma_\mu^2\mathcal{T}} + \frac{\sigma_\mu^4\mathcal{T}^2}{(\sigma_W^2 + \sigma_\mu^2\mathcal{T})^2}\right)\\ &= \frac{1}{2\sigma_W^4}\left(n - 1 + \left(1 - \frac{\sigma_\mu^2\mathcal{T}}{\sigma_W^2 + \sigma_\mu^2\mathcal{T}}\right)^2\right) = \frac{n-1}{2\sigma_W^4} + \frac{\omega^2}{2\mathcal{T}^2}.\end{aligned} \tag{A1}$$

To prove (11), notice that

$$\det\mathscr{M} = \mathscr{M}_{11}(\mathscr{M}_{22}\mathscr{M}_{33} - \mathscr{M}_{23}^2) = \frac{(n-1)\mathscr{M}_{11}^3}{4\sigma_W^4}. \tag{A2}$$

The inverse of $\mathscr{M}$ is

$$\begin{aligned}\mathbf{D} = \mathscr{M}^{-1} &= \frac{1}{\det\mathscr{M}}\begin{pmatrix}\mathscr{M}_{22}\mathscr{M}_{33} - \mathscr{M}_{23}^2 & 0 & 0\\ 0 & \mathscr{M}_{11}\mathscr{M}_{33} & -\mathscr{M}_{11}\mathscr{M}_{23}\\ 0 & -\mathscr{M}_{11}\mathscr{M}_{23} & \mathscr{M}_{11}\mathscr{M}_{22}\end{pmatrix}\\ &= \begin{pmatrix}\omega^{-1} & 0 & 0\\ 0 & \frac{2\sigma_W^4}{n-1} & -\frac{2\sigma_W^4}{(n-1)\mathcal{T}}\\ 0 & -\frac{2\sigma_W^4}{(n-1)\mathcal{T}} & 2\omega^{-2} + \frac{2\sigma_W^4}{(n-1)\mathcal{T}^2}\end{pmatrix}.\end{aligned}$$

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